Notes on the Little-Parks Effect in a bottlenecked nanowire

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I. STATEMENT OF THE PROBLEM

In these notes we study a doubly-connected superconducting cylinder narrowing along the longitudinal direction and threaded by a magnetic flux. We assume that the nanoshell thickness $d(z) \equiv R_2(z) - R_1(z)$ is much smaller than the London penetration length, therefore, the screening (Meissner effect) is negligible and the flux is not quantized. Further, we consider a superconductor in the dirty limit (i.e. the mean free path is the smallest length scale) and describe it within the quasiclassical Usadel formalism.

Noting that the winding number n in the phase of the superconducting order parameter takes integer values and adapts to minimize the free energy for a given magnetic flux, the bottlenecked geometry of the nanowire raises the question of whether n can be discontinuous along the longitudinal direction. Here, we address this question for a sharp bottlenecked (Fig. 1).

We parametrize the inner ($i = 1$) and outer ($i = 2$) radii of the cylinder as $R_i(z) = R_{i,+}\theta(z) + R_{i,-}\theta(-z)$, with $\theta(z)$ the Heaviside theta function.

Further, we assume that $\xi_0 \gg R_{i,\pm}$ and $\xi_0 \ll \ell$, with ξ_0 the superconducting coherence length at zero magnetic flux and ℓ the half-length of the cylinder.

The magnetic field is set along the z direction, $\boldsymbol{h} = h\hat{\boldsymbol{z}}$, and we choose the Coulomb gauge, $\boldsymbol{A} = \frac{1}{2}hr\hat{\boldsymbol{\varphi}}$ with r the radial coordinate.

In the rest of the notes we set $\hbar = c = k_{\mathrm{B}} = 1$ and $e > 0$ the electron charge.

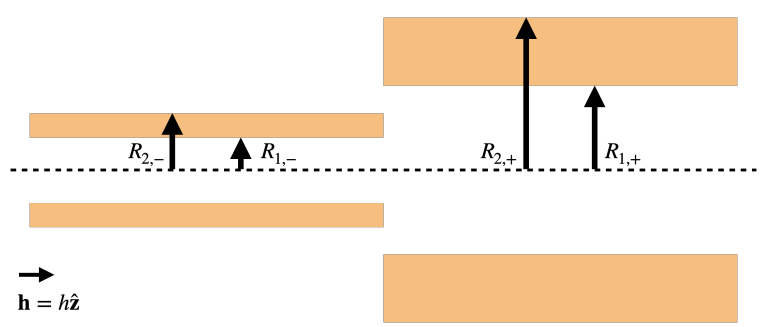

FIG. 1. Schematic view of the nanowire. Although the sharp bottleneck at $z = 0$ is unphysical, it should be understood as the limiting case of a connected bottleneck with characteristic length $\lambda_{\mathrm{BN}} \to 0$.

II. CALCULATION OF THE ANOMALOUS GREEN FUNCTION

Our starting point is the linearized Usadel equation for the anomalous component of the momentum-averaged quasiclassical Green function f,

$$-D(\boldsymbol{\nabla} - 2ei\boldsymbol{A})^2 f + 2|\omega| f + 2i\Delta = 0, \tag{1}$$

and the boundary condition $\boldsymbol{\nabla}_\perp f|_{\text{bound.}} = 0$.

Owing to the "very small" nanoshell thickness, we assume that f is finite and constant inside the nanoshell along the radial direction, but vanishes outside[1]. Therefore, we can radially integrate Eq. (1) to obtain,

$$-D\left\{\frac{1}{R_2^2(z) - R_1^2(z)} \log\left(\frac{R_2(z)}{R_1(z)}\right)\partial_\varphi^2 - eih\partial_\varphi + \frac{1}{2}\partial_z^2 - \left[\left(R_2^2(z) - R_1^2(z)\right)\left(\frac{eh}{2}\right)^2 + \frac{|\omega|}{D}\right]\right\} f + i\Delta = 0 \tag{2}$$

We consider the following ansätze for the anomalous part of the Green function f and the order parameter Δ,

$$f(z,\varphi) = \sum_{a,j} f_{a,j}(z) e^{in_a\varphi}\theta(jz), \tag{3a}$$

$$\Delta(z,\varphi) = \sum_{a,j} |\Delta_{a,j}(z)| e^{in_a\varphi}\theta(jz), \tag{3b}$$

with φ the angular coordinate. The running index $j = \{+,-\}$ denotes the right ($z > 0$) and left ($z < 0$) parts of the cylinder, while $a = \{\mathrm{R}, \mathrm{L}\}$ indicates the intrinsic winding number on each half, n_R and n_L. The crucial aspect of this ansatz is that it allows for a superposition of both winding numbers on a given half, thereby yielding a φ-dependent global amplitude.

To solve Eq.(2), we omit the self-consistent treatment of the order parameter and drop the z dependence in Eq. (3b). Then, we exploit the linearity of the resulting second order ODE to write four one-dimensional differential equations,

$$\left[-\frac{D}{2}\partial_{zz}^2 + \Omega_{a,j}\right] f_{a,j}(z) + i|\Delta_{a,j}| = 0, \tag{4}$$

where $\Omega_{a,j} = |\omega| + \alpha_{a,j}$, and

$$\alpha_{a,j} = D\left\{\left(\frac{eh}{2}\right)^2 \left[R_{2,j}^2 + R_{1,j}^2\right] - ehn_a + \frac{1}{R_{2,j}^2 - R_{1,j}^2}\log\left(\frac{R_{2,\pm}}{R_{1,j}}\right) n_a^2\right\} \tag{5}$$

[1] In the following, this trivial r dependence is omitted to lighten the notation.

is the depairing parameter. Each differential equation is supplemented by the usual boundary conditions ensuring continuity and differentiability at the junction and convergence to the bulk value at the boundaries of the cylinder,

$$f_{a,+}(0) = f_{a,-}(0), \tag{6a}$$

$$\partial_z f_{a,+}(0) = \partial_z f_{a,-}(0), \tag{6b}$$

$$f_{a,j}(z \to j\infty) \equiv f^{\mathrm{o}}_{a,j}. \tag{6c}$$

In total, we have

$$f_{a,j}(z) = f^{\mathrm{o}}_{a,j}\left(1 - \chi_{a,j} e^{-jz/\xi_{a,j}}\right), \tag{7}$$

with

$$f^{\mathrm{o}}_{a,j} = -\frac{i|\Delta_{a,j}|}{\Omega_{j,a}}, \quad \xi_{a,j} = \sqrt{\frac{D}{2\Omega_{a,j}}}, \quad \chi_{a,j} = \left(1 - \frac{f^{\mathrm{o}}_{a,\bar{j}}}{f^{\mathrm{o}}_{a,j}}\right)\frac{\xi_{a,j}}{\xi_{a,j} + \xi_{a,\bar{j}}}. \tag{8}$$

It is worth noting that $f_{a,j}(z)$ on a given half cylinder depends on the winding number n_a of said half only, but involves terms with the two radii $j = \pm$ through the factors $\chi_{a,j}$. The latter naturally follows from fulfilling the continuity and differentiability conditions.

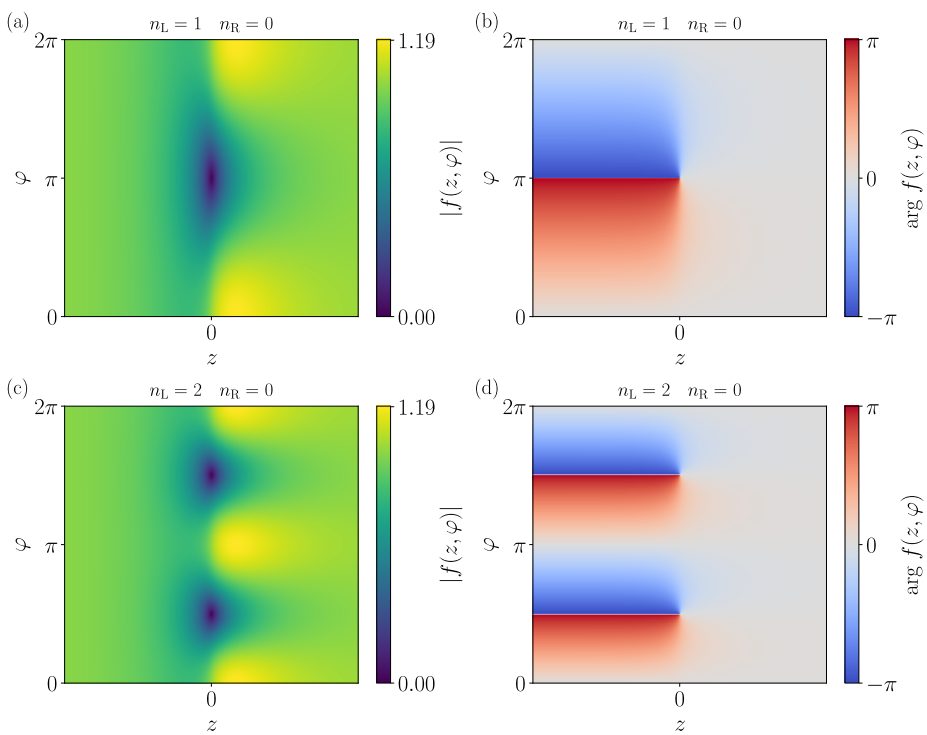

FIG. 2. Comparison of $f(z,\varphi)$ for $n_{\mathrm{L}} - n_{\mathrm{R}} = 1$ (first row) and $n_{\mathrm{L}} - n_{\mathrm{R}} = 2$ (second row). Parameters: $f^{\mathrm{o}}_{R,+} = f^{\mathrm{o}}_{L,-} = 1$, $f^{\mathrm{o}}_{R,-} = f^{\mathrm{o}}_{L,+} = 0$, $\xi_{R,+} = \xi_{R,-} = 0.1$, $\xi_{L,+} = \xi_{L,-} = 0.3$.

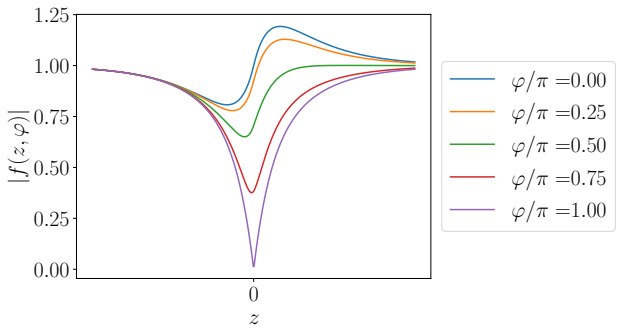

FIG. 3. Profile of $|f(z,\varphi)|$ for various angles φ for $n_{\rm L}-n_{\rm R}=1$ [Fig. 2 (a)]

The solution for $f(z,\varphi)$ exhibits a line with $|n_{\rm L}-n_{\rm R}|$ vortices where $|f(z_{\rm c},\varphi_{\rm c})|=0$ (Fig. 2). However, the position of the vortices $(z_{\rm c},\varphi_{\rm c})$ does not necessarily coincide with the position of the bottleneck (here, $z=0$), but it is given by

$$f_{\rm R}(z_{\rm c})=f_{\rm L}(z_{\rm c}), \tag{9a}$$

$$\cos\left[(n_{\rm L}-n_{\rm R})\varphi_{\rm c}\right]=-1, \tag{9b}$$

with $f_a(z)=f_{a,+}(z)\theta(z)+f_{a,-}(z)\theta(-z)$.

We note that in Figs. 2 and 3, the parameters $f^{\rm o}_{a,j}$ and $\xi_{a,j}$ are taken as degrees of freedom for illustration purposes, but in reality they depend on the depairing parameter $\alpha_{a,j}$ (and $|\omega|$). Enforcing $n_{\rm L}-n_{\rm R}=1$, imposes a constraint on $\alpha_{a,j}$, namely, the solution with $n_{\rm L}-n_{\rm R}=1$ must minimize the free energy (see Sec. III). In Fig. 4 we show an example with "physically-consistent" parameters. The z-profile of the solutions slightly depends on this choice, but the vortex structure remains.

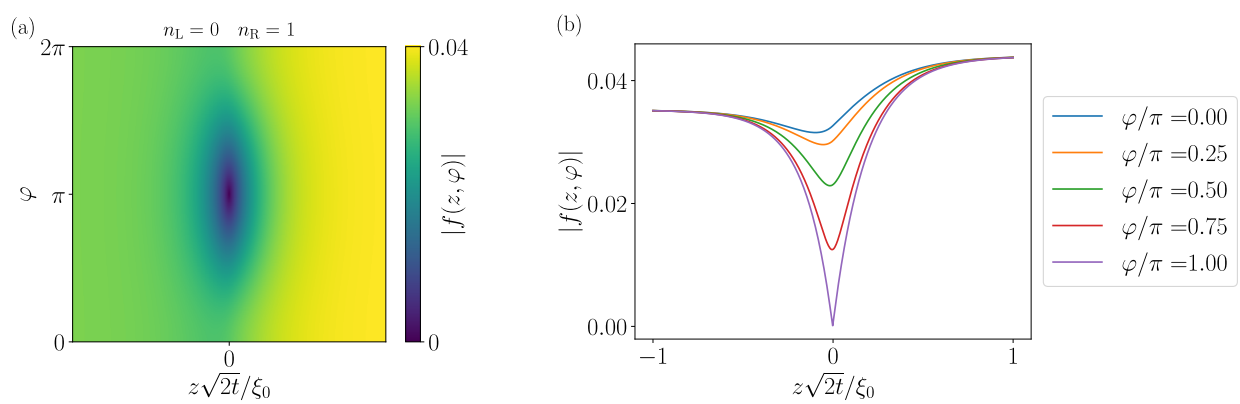

FIG. 4. Absolute value of the anomalous Green function $|f(z,\varphi)|$ at $\omega=0$ in a parameter regime such that $n_{\rm L}-n_{\rm R}=-1$ minimizes the total free energy. Parameters: $\overline{\phi}=0.7$, $\lambda_{\overline{R}_{\rm T}}=10$, $\epsilon=0.1$, $\overline{R}_-/\overline{R}_+=0.5$, $t=0.5$, $|\Delta|/(\pi T)=1$. [See Eq. (8) for the parameter definitions].

III. FREE ENERGY

Now we proceed to compute the free energy for the solution of the anomalous Green function found in the previous section. The goal is to determine under which conditions $n_{\rm L} \neq n_{\rm R}$ is the most energetically favorable scenario.

Up to Δ-independent constant terms, the free energy reads

$$F_s = -\frac{\pi T \nu_0}{2} \sum_{\omega} \int_V d\boldsymbol{r} \left\{ i\left(\Delta \overline{f} + \Delta^* f\right) + \frac{D}{2}\left[\boldsymbol{\nabla} f \boldsymbol{\nabla} \overline{f} - 2ei\boldsymbol{A}\left(f\boldsymbol{\nabla}\overline{f} - \overline{f}\boldsymbol{\nabla} f\right) + (2e\boldsymbol{A})^2 f\overline{f}\right]\right\}, \quad (10)$$

where the integral is taken over the volume of the nanowire, "$*$" denotes complex conjugation and $\overline{f}(\varphi) = f(-\varphi)$. In total we have,

$$F_s = -\frac{\nu_0 \overline{V}}{2}\left(F_{\rm A} + F_{\rm B} + F_{\rm C}\right), \quad (11)$$

where

$$F_{\rm A} = \sum_{a,b,j} \epsilon_j \sqrt{\gamma_j}\delta_{n_a,n_b}|\Delta_{a,j}||\Delta_{b,j}|\left[S^{(1)}_{a,j} - \frac{1}{4}\widetilde{\alpha}_{a,j} S^{(2)}_{a,b,j} - \frac{\lambda_\ell}{\sqrt{2t}}\left(S^{(3/2)}_{a,b,j} - \frac{1}{4}\widetilde{\alpha}_{a,j} S^{(5/2)}_{a,b,j}\right)\right], \quad (12a)$$

$$F_{\rm B} = \frac{\lambda_\ell}{\sqrt{2t}} \sum_{a,b,j} \epsilon_j \sqrt{\gamma_j}\delta_{n_a,n_b}|\Delta_{a,j}||\Delta_{b,\overline{j}}|\left(T^{(3/2)}_{a,b,j} - \frac{1}{4}\widetilde{\alpha}_{a,j} T^{(5/2)}_{a,b,j}\right), \quad (12b)$$

$$F_{\rm C} = -\frac{\lambda_\ell}{\sqrt{2t}} \sum_{a,b,j} \epsilon_j \sqrt{\gamma_j}\delta_{n_a,n_b}|\Delta_{a,\overline{j}}||\Delta_{b,\overline{j}}|\left(Q^{(3/2)}_{a,b,j} + \frac{1}{4}\widetilde{\alpha}_{a,j} Q^{(5/2)}_{a,b,j}\right), \quad (12c)$$

where $S^{(\nu)}_{a,b,j}, T^{(\nu)}_{a,b,j}, Q^{(\nu)}_{a,b,j}$ are positive and dimensionless functions of $\widetilde{\alpha}_{a,j}$ defined in App. A 1 and δ_{n_a,n_b} is the Kronecker delta. In addition, we introduced a dimensionless depairing parameter,

$$\widetilde{\alpha}_{a,j} = \frac{\lambda^2_{\overline{R}_{\rm T}}}{t}\frac{1}{\gamma_j}\left[\left(\overline{\phi}\gamma_j - n_a\right)^2 + \frac{1}{\gamma_j}\frac{\epsilon_j^2}{4}\left(\overline{\phi}^2\gamma_j^2 + \frac{n_a^2}{3}\right)\right], \quad (13)$$

and the following parameters,

$$\gamma_j = \sqrt{\frac{\overline{R}_j}{\overline{R}_+ + \overline{R}_-}}, \quad \epsilon_j = \frac{d_j}{\overline{R}_{\rm T}}, \quad \lambda_\ell = \frac{\xi_0}{\ell}, \quad \lambda_{\overline{R}_{\rm T}} = \frac{\xi_0}{\overline{R}_{\rm T}}, \quad t = \frac{T}{T^0_{\rm c}}, \quad \overline{\phi} = \frac{\overline{\Phi}}{\Phi_0}, \quad (14)$$

where $\overline{R}_j = \frac{R_{1,j}+R_{2,j}}{2}$ and $d_j = R_{2,j} - R_{1,j}$ are the nanoshell's average radius and thickness at the $j = \pm$ half-plane, $\overline{R}_{\rm T} = \overline{R}_+ + \overline{R}_-$ yields the "average" flux $\overline{\Phi} = h\pi\overline{R}^2_{\rm T}$ and controls the total-volume prefactor $\overline{V} = 2\pi\ell\overline{R}^2_{\rm T}$ in Eq. (11), and $\xi_0 = \sqrt{\frac{D}{2\pi T^0_{\rm c}}}$ is proportional to the superconducting (diffusive) coherence length at zero magnetic field.

We note that in obtaining the expressions of the free energy in Eqs. (12) we assumed $\xi_{a,j}/\ell \ll 1$ and therefore neglected terms proportional to $\sum_\omega e^{-\ell/\xi_{a,j}}$. Further, we expanded the depairing parameter in Eq. (5) to third order in $\frac{d_j}{\overline{R}_j} \ll 1$.

Next, we analyze the free energy as a function of the average flux $\overline{\phi}$ for a non-unity $\overline{R}_-/\overline{R}_+$ ratio. For simplicity we set $\epsilon_+ = \epsilon_- \equiv \epsilon$ and we fix $t = 0.5$. The parameter regime consistent with our assumptions reads as follows

$$\epsilon \ll 1, \quad \lambda_\ell \ll 1, \quad \lambda_{\overline{R}_\mathrm{T}} \gg 1. \tag{15}$$

We further simplify by assuming $|\Delta_{\mathrm{R},+}| = |\Delta_{\mathrm{L},-}| = |\Delta|$ and $|\Delta_{\mathrm{R},-}| = |\Delta_{\mathrm{L},+}| = 0$. This can be understood as the first iteration in the self-consistent procedure, where the order parameter on a given sector of the cylinder only depends on the intrinsic winding number of that sector. Note that this assumption does not yield vanishing $f_{\mathrm{R},-}(z), f_{\mathrm{L},+}(z)$ owing to the mixing factor $\chi_{a,j}$ [see Eq. (8)]. The sums in Eqs. (12) contain then the following terms:

$$F_\mathrm{A} : \sum_{a,b,j} \longrightarrow (\mathrm{R},\mathrm{R},+),\ (\mathrm{L},\mathrm{L},-), \tag{16a}$$

$$F_\mathrm{B} : \sum_{a,b,j} \longrightarrow (\mathrm{R},\mathrm{L},+),\ (\mathrm{L},\mathrm{R},-), \tag{16b}$$

$$F_\mathrm{C} : \sum_{a,b,j} \longrightarrow (\mathrm{R},\mathrm{R},-),\ (\mathrm{L},\mathrm{L},+). \tag{16c}$$

Let us start the analysis by concentrating on F_A. The first term in Eq. (12a) represents the energy gain due to the superconductivity condensation, the second term is a penalization due to the applied magnetic field and the third and fourth terms are a penalization due to spatial variation of f in the vicinity of the bottleneck. Naturally, the second term vanishes at zero depairing parameter, and third and fourth terms do so at $L \to \infty$.

It follows from Eq. (16b) that under our assumptions F_A is simply the sum of the free energy from each half cylinder. This simplification renders the analysis straightforward: the optimal $n_\mathrm{R}, n_\mathrm{L}$ are chosen independently, so that they minimize their corresponding contribution to F_A. For any $\overline{R}_-/\overline{R}_+ \neq 1$ there will exist a set of intervals of the magnetic flux where $n_\mathrm{L} \neq n_\mathrm{R}$ minimizes the total F_A, which will grow in size as $\overline{\phi}$ increases. In particular, if $\overline{R}_-/\overline{R}_+ < 1$, then the transition to the next winding number happens earlier on the $+$ side than on the $-$ side because there is a larger flux through the cylinder on the $+$ side than on the $-$ side. Therefore, we have that $\Delta n \equiv n_\mathrm{R} - n_- = 1$ for $\overline{\phi} \in [\overline{\phi}_{\mathrm{c},+}(n), \overline{\phi}_{\mathrm{c},-}(n)]$ where n is an arbitrary integer and $\overline{\phi}_{\mathrm{c},j}(n)$ is the critical flux at which the optimal winding number n increases by one on the j side of the cylinder. This is given by the equation $\widetilde{\alpha}_{a,j}(\overline{\phi}_{\mathrm{c},j}, n) = \widetilde{\alpha}_{a,j}(\overline{\phi}_{\mathrm{c},j}, n+1)$, and following Eq. (13) it reads

$$\overline{\phi}_{\mathrm{c},j}(n) = \frac{(1+2n)(\epsilon^2 + 12\gamma_j)}{24\gamma_j^2}. \tag{17}$$

The analysis of Eq. (17) corroborates the physical intuition that the size of said intervals $\Delta\overline{\phi}_{\Delta n=1} \equiv \overline{\phi}_{\mathrm{c},-}(n) - \overline{\phi}_{\mathrm{c},+}(n)$ grows with n and the nanoshell thickness ϵ, and decreasing $\overline{R}_-/\overline{R}_+ < 1$. Further, we note that for sufficiently large n (or equivalently, $\overline{\phi}$), ϵ, or sufficiently small $\overline{R}_-/\overline{R}_+$, it is possible to have $\Delta n > 1$. In particular, $\Delta n \geq 2$ if $\overline{\phi}_{\mathrm{c},+}(n+1) < \overline{\phi}_{\mathrm{c},-}(n)$, etc.

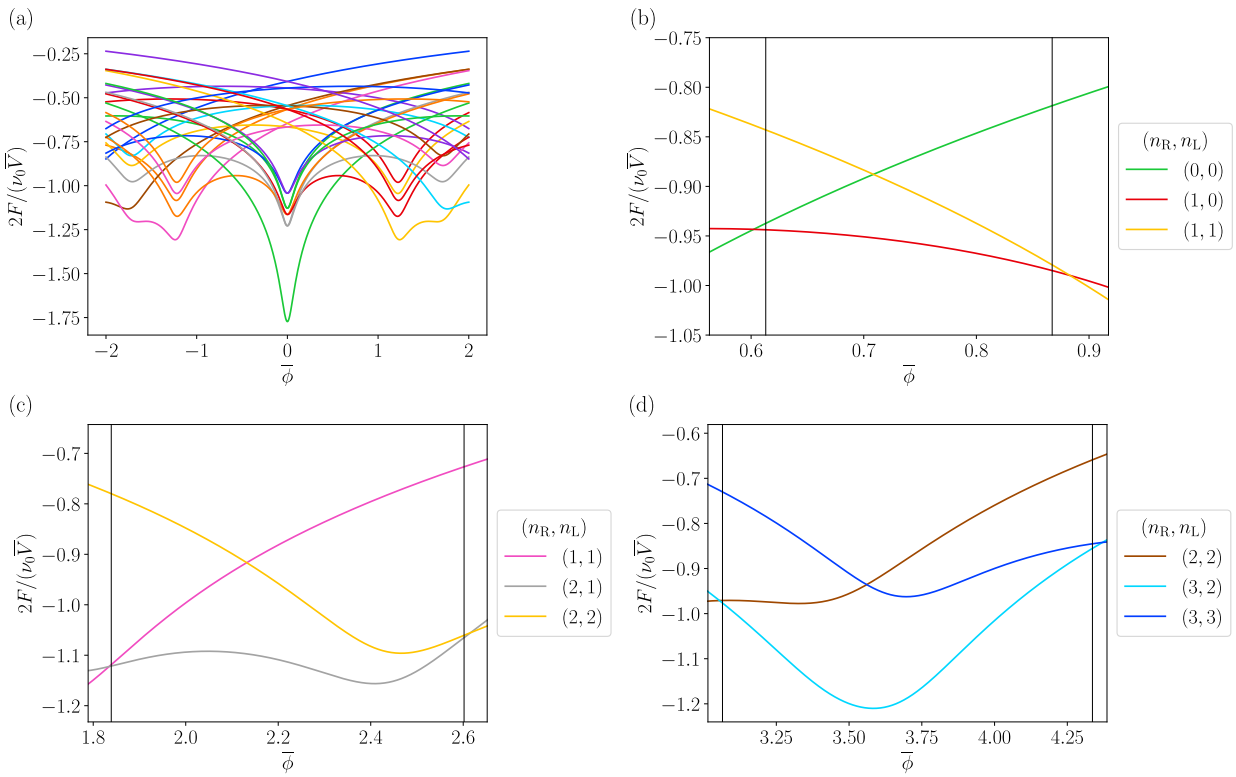

FIG. 5. Analysis of the free energy. (a) General plot for various values of $n_\mathrm{L}, n_\mathrm{R}$ as a function of the average magnetic flux. (b,c,d) Free energy in three different $\overline{\phi}$-intervals where $\Delta n = 1$ is the preferred scenario. The vertical lines are given by Eq. (17) and indicate the n-switching point in the limit of an infinite cylinder. Calculation parameters: $\lambda_{\overline{R}_\mathrm{T}} = 10$, $\lambda_\ell = 0.1$, $\epsilon = 0.1$, $\overline{R}_-/\overline{R}_+ = 0.5$, $|\Delta| = 1$, $n_\mathrm{D} = 10^4$.

If we take into account the terms $F_\mathrm{B}, F_\mathrm{C}$, the analysis becomes more complicated since they contain mixed terms (i.e., the depairing parameter involves the radii of one cylinder half and the intrinsic winding number of the other half). However, we note that both $F_\mathrm{B}, F_\mathrm{C}$ are proportional to $\lambda_\ell \ll 1$, so we can expect a $n_\mathrm{L} \neq n_\mathrm{R}$ to be preferred in general, with small corrections on the criterion given in Eq. (17). This has a clear physical interpretation: in the limit of a very long nanoshell, the free energy is essentially controlled by the bulk parameters and the corrections to the free energy due to the vortices in $f(z, \varphi)$ at the bottleneck become negligible.

IV. CHARGE CURRENT

The charge current reads

$$\boldsymbol{j}(\boldsymbol{r}) = -i\frac{\pi T \nu_0 D e}{2} \sum_{\omega} \left(f \boldsymbol{\nabla} \overline{f} - \overline{f} \boldsymbol{\nabla} f + 4ei\boldsymbol{A} f \overline{f} \right) . \tag{18}$$

In total, the longitudinal and angular components read

$$j_z(z,\varphi) = 2e\nu_0 \frac{\xi_0}{\sqrt{2t}} \sum_{a,b,j} P^{(z)}_{a,b,j}(\widetilde{z}) \sin\left[(n_a - n_b)\varphi\right] , \tag{19a}$$

$$j_\varphi(r,z,\varphi) = 2e\nu_0 \frac{\xi_0}{\sqrt{2t}} \frac{\lambda_{\overline{R}_\mathrm{T}}}{\sqrt{2t}} \sum_{a,b,j} P^{(\varphi)}_{a,b,j}(\widetilde{z}) \cos\left[(n_a - n_b)\varphi\right] \frac{1}{\widetilde{r}} \left(n_a - \overline{\phi}\widetilde{r}^2 \right) , \tag{19b}$$

where $\widetilde{r} = \frac{r}{\overline{R}_\mathrm{T}}$, $\widetilde{z} = z\frac{\sqrt{2t}}{\xi_0}$ and $P^{(z,\varphi)}_{a,b,j}(\widetilde{z})$ have units of $|\Delta|^2$ and exponentially decay away from $z = 0$ (see App. A 2). The analysis of Eq. (19a) shows that the maximum amplitude of the longitudinal component is always at $z = 0$, and the number and position of its zeros is given by $\sin\left[(n_\mathrm{L} - n_\mathrm{R})\varphi\right] = 0$ [Fig. 6 (a)].

The charge current in the (z,φ) plane exhibits a vortex which naturally coincides with the vortex in the anomalous Green function [compare Fig. 4 (a) and Fig. 7].

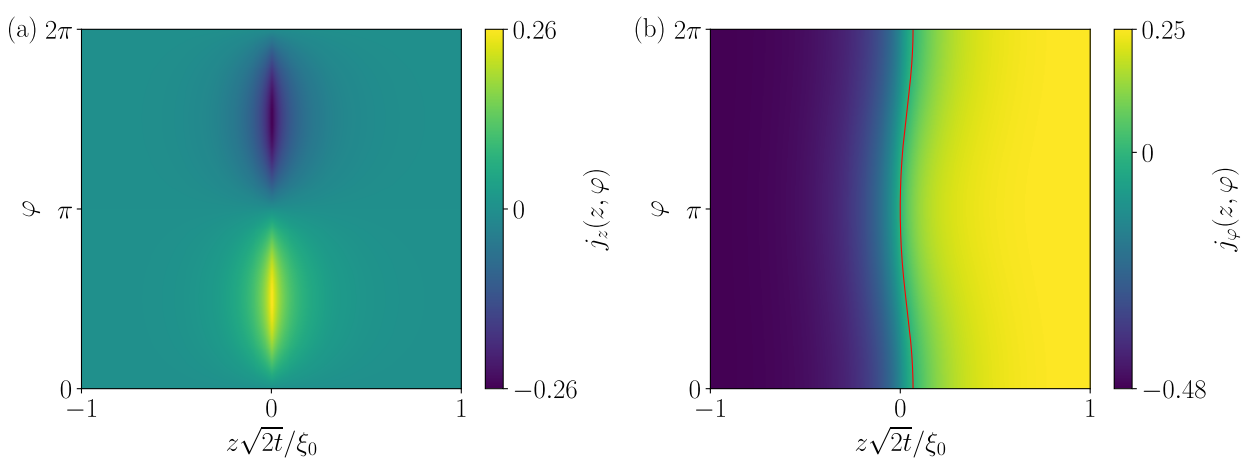

FIG. 6. Charge current for $n_\mathrm{R} = 1$, $n_\mathrm{L} = 0$. (a) Longitudinal component. It vanishes at $\varphi = 0, \pi$. (b) Azimuthal component. It vanishes along the red line. Parameters: Same as Fig. 2 with $\overline{\phi} = 0.7$, $\widetilde{r} = 1$. Color bar in units of $2e\nu_0|\Delta|^2\frac{\xi_0}{\sqrt{2t}}$.

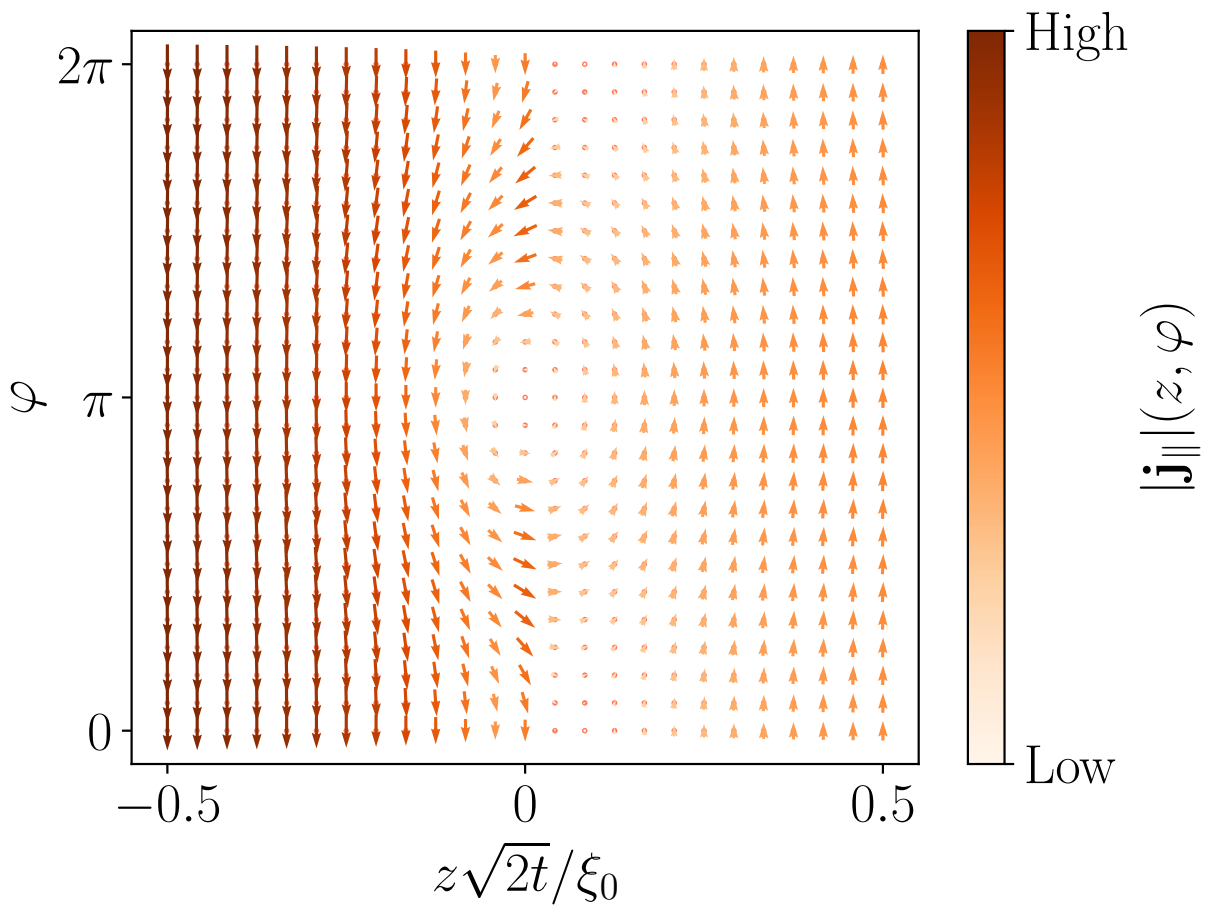

FIG. 7. Vector plot of the charge current combining j_z and j_φ in Fig. 6 with a zoom on the z axis. A vortex-antivortex pair appears close to $\widetilde{z} = 0$ and $\varphi = \pi, 0$.

Appendix A: Details on the analytical expressions

1. Free energy

We introduce the following variables to lighten the notation,

$$X = \widetilde{\Omega}_{a,j}, \quad Y = \widetilde{\Omega}_{a,\bar{j}}, \quad Z = \widetilde{\Omega}_{b,j}, \quad W = \widetilde{\Omega}_{b,\bar{j}}, \tag{A1}$$

with $\widetilde{\Omega}_{a,j} = n + \frac{1+\widetilde{\alpha}_{a,j}}{2}$. Then, the variables in Eqs.(12) read

$$S^{(1)}_{a,j} = \sum_{n=0}^{\infty} \frac{1}{X}, \tag{A2a}$$

$$S^{(2)}_{a,b,j} = \sum_{n=0}^{\infty} \frac{1}{XZ}, \tag{A2b}$$

$$S^{(3/2)}_{a,b,j} = \frac{1}{2} \sum_{n=0}^{\infty} \Gamma \sqrt{\frac{Y}{X^3 Z}} \left[2Z \left(\sqrt{X} + \sqrt{Z} + \sqrt{W} \right) + \sqrt{W} X + 2\sqrt{XZW} \right], \tag{A2c}$$

$$T^{(3/2)}_{a,b,j} = \sum_{n=0}^{\infty} \Gamma \frac{X \left(\sqrt{X} + \sqrt{Y} + \sqrt{Z} \right) + Z\sqrt{Y} + \sqrt{XYZ}}{\sqrt{XZW}}, \tag{A2d}$$

$$Q^{(3/2)}_{a,b,j} = \frac{1}{2} \sum_{n=0}^{\infty} \Gamma \sqrt{\frac{XZ}{YW}}, \tag{A2e}$$

$$S^{(5/2)}_{a,b,j} = \sum_{n=0}^{\infty} \Gamma \frac{1}{(XZ)^{3/2}} \left[X\sqrt{W} \left(\sqrt{X} + \sqrt{Y} + \sqrt{Z} \right) + Z\sqrt{Y} \left(\sqrt{X} + \sqrt{Z} + \sqrt{W} \right) + \sqrt{XYZW} \right], \tag{A2f}$$

$$T^{(5/2)}_{a,b,j} = 2 \sum_{n=0}^{\infty} \Gamma \frac{\sqrt{X} + \sqrt{Y} + \sqrt{Z}}{\sqrt{XZW}}, \tag{A2g}$$

$$Q^{(5/2)}_{a,b,j} = \sum_{n=0}^{\infty} \Gamma \frac{1}{\sqrt{YW}}, \tag{A2h}$$

where

$$\Gamma = \frac{1}{\left(\sqrt{X} + \sqrt{Y} \right) \left(\sqrt{X} + \sqrt{Z} \right) \left(\sqrt{Z} + \sqrt{W} \right)}. \tag{A3}$$

All the expressions above are bound except $S^{(1)}_{a,b,j}$ for which we cutoff the infinity sum to $n_{\mathrm{D}} \approx \frac{\omega_{\mathrm{D}}}{2\pi T}$, with ω_{D} the Debye frequency. In practice we set $n_{\mathrm{D}} = 10^4$ in all sums.

2. Charge current

The functions of $\widetilde{z}$ introduced in Eq. (19b) read,

$$P^{(z)}_{a,b,j}(\widetilde{z}) = \sum_{n=0}^{\infty} j\ \theta\left(j\widetilde{z}\right)\left(\frac{|\Delta_{a,j}|}{X} - \frac{|\Delta_{a,\bar{j}}|}{Y}\right)\frac{\sqrt{XY}}{\sqrt{X}+\sqrt{Y}}\cdot \tag{A4a}$$
$$\cdot\left[\frac{|\Delta_{b,j}|}{Z} - \left(\frac{|\Delta_{b,j}|}{Z} - \frac{|\Delta_{b,\bar{j}}|}{W}\right)\frac{\sqrt{W}}{\sqrt{Z}+\sqrt{W}}e^{-|\widetilde{z}|\sqrt{Z}}\right]e^{-|\widetilde{z}|\sqrt{X}},$$

$$P^{(\varphi)}_{a,b,j}(\widetilde{z}) = \sum_{n=0}^{\infty} \theta\left(j\widetilde{z}\right)\left[\frac{|\Delta_{a,j}|}{X} - \left(\frac{|\Delta_{a,j}|}{X} - \frac{|\Delta_{a,\bar{j}}|}{Y}\right)\frac{\sqrt{Y}}{\sqrt{X}+\sqrt{Y}}e^{-|\widetilde{z}|\sqrt{X}}\right]\cdot \tag{A4b}$$
$$\cdot\left[\frac{|\Delta_{b,j}|}{Z} - \left(\frac{|\Delta_{b,j}|}{Z} - \frac{|\Delta_{b,\bar{j}}|}{W}\right)\frac{\sqrt{W}}{\sqrt{Z}+\sqrt{W}}e^{-|\widetilde{z}|\sqrt{Z}}\right],$$

where X, Y, W, Z were introduced in Eqs. (A1). The expressions in Eq. (A4) could be rendered more synthetic, but under its current form it is clear that $P^{(z,\varphi)}_{a,b,j}(\widetilde{z})$ is maximal at $\widetilde{z} = 0$, and vanishes at infinity.